Supplementary material: benchmark documentation

Done Is Not Correct: Measuring Silent Failures and Self-Verification Calibration When LLM Agents Take CAD Actions

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1 Access for reviewers

Repository: `https://github.com/rodriguescarson/cad-silent-failure-bench`. Archived, versioned copy: `https://doi.org/10.5281/zenodo.22286324` (suite v1.0, the version every number in the paper is computed from). The repository README documents the layout; `scripts/regrade.py` regrades every stored attempt offline in about twenty minutes on a laptop, `scripts/stats.py` produces every statistic in the paper from the stored records without any API call, and `scripts/tolerance_sensitivi` produces the band-scaling table. A static 3D viewer (`site/`) renders every attempt against its reference solution and needs only a local HTTP server.

2 Author statement and licence

The authors bear all responsibility in case of violation of rights. The released material was created by the authors: task specifications and reference solutions were written by the authors, and attempt records were generated by the authors through the model vendors' public APIs under those vendors' terms of use, which permit publication of outputs for research. It contains no personal data and no data about people; the only human judgement it records is a co-author's blind pass/fail scoring of 81 rendered parts. Code is released under the MIT licence. Task specifications, reference solutions, attempt records, transcripts, measured properties, the expert scoring sheet, and the viewer assets are released under CC BY 4.0. Both licence texts are in the repository root.

3 Hosting, licensing, and maintenance plan

- **Hosting.** A public GitHub repository holds the living suite and code. A Zenodo record holds the frozen v1.0 archive cited in the paper; Zenodo records are immutable and DOI-resolvable independently of GitHub.
- **Versioning.** v1.0 is the paper's version and will not change. Later suite versions (roadmap items: dual phrasings, parametric re-instantiation, an assembly tier, ISO 2768 band policy, the failure taxonomy) are tagged releases with their own Zenodo DOIs, so results on different versions are never conflated.
- **Corrections.** Any correction to a task, check, or record is logged in `ERRATA.md` with the date and effect on the paper's numbers; nothing is changed silently.

- **Maintenance.** The first author maintains the repository, answers issues, and keeps the release runnable against the pinned kernel version (`build123d` 0.11, OpenCascade via OCP) for at least three years from publication. The grader and harness are pure Python with the kernel isolated in one module, so kernel upgrades are contained.

- **Contact.** Issues on the repository, or carson@celabe.com.

4 Datasheet

Structured after Gebru et al. (2021), *Datasheets for datasets.*

Motivation

For what purpose was the dataset created? To measure, under an exact geometric oracle, how often LLM agents that write parametric CAD produce a valid, renderable, wrong part while claiming completion, and how well their stated confidence tracks correctness; and to compare agent architectures on that measurand. **Who created it and who funded it?** The three authors; no funding.

Composition

What do the instances represent? Two instance types. (a) *Tasks*: 13 natural-language mechanical specification tasks in three tiers (5 single-feature, 6 multi-feature, 2 edit-under-constraint), each with a list of machine-checkable requirements (122 checks in total: 58 dimensional `approx`, 13 `range`, 13 `equal`, 13 validity, 22 `cylinder_count` with axial-extent filters, 3 `bolt_circle`) and a reference solution. (b) *Attempts*: one agent attempt at one task under one condition, as a JSON record. **How many instances?** 13 tasks; 519 attempts in the full population (four models with complete 13-task coverage, three repetitions, three conditions, plus a partial fifth model), of which 468 form the primary population; 156 ablation attempts (legacy claim contract, all four models); 60 pilot attempts. Multi-turn transcripts for 312 attempts (156 fixed-protocol tool-use, 156 ablation). **What does each attempt record contain?** Task id, tier, model, condition, repetition, final code, measured-properties dictionary (bounding box, volume, centre of mass, solid/face/edge counts, cylindrical features with radius, axis, and axial extent), pass/fail verdict under the original and the hardened oracle, list of failed checks, completion claim, stated confidence, step count, input and output tokens, error text if any, and per-round context-divergence scores for multi-agent episodes. **Is any information missing?** Transcripts were not persisted for single-shot and multi-agent attempts in the full population (final code and all measurements were); the fifth model (gpt-5.5) covers an easy-tier prefix only, by API budget. **Does it contain confidential or personal data?** No. **Are there recommended splits?** None; it is an evaluation suite. The reference solutions must not be used to train models that are then evaluated on the suite.

Collection process

How was the data acquired? Specifications were authored by the authors, one a practising mechanical engineer, with every requirement mapped to a check before any agent run; reference solutions were written and passed through a calibration gate requiring each to satisfy every check of its task. Attempts were generated in June 2026 through the Anthropic API and OpenRouter at provider-default decoding, 4,096 output tokens, three repetitions per cell, tool-use and multi-agent episodes capped at four steps, kernel execution capped at 60 s in an isolated subprocess. **Who was**

involved and were they compensated? The authors only. **Ethical review?** Not applicable; no human subjects.

Preprocessing, cleaning, labelling

Every stored attempt was re-executed and regraded under the hardened oracle after a mid-study audit; both verdicts are retained per record and the flips are reported in the paper (18 pass to fail, 0 fail to pass). A stratified sample of 81 valid-solid outputs was blind-scored by a co-author against the specifications; the sheet and the κ computation are released.

Uses

Has it been used for any tasks already? The paper's analyses. **What other tasks could it be used for?** Auditing grader leniency in CAD benchmarks; studying verbalized-confidence calibration in agents; training or evaluating verifiers on the transcripts; failure-mode taxonomy work. **Are there uses it should not be used for?** It should not be read as certifying any generated part: the checks are model-correctness screens, not manufacturability or functional validation. Absolute rates are lower bounds on one 13-task suite.

Distribution

Public, under MIT (code) and CC BY 4.0 (everything else), via GitHub and Zenodo; no fees, no access restrictions, no export controls.

Maintenance

See Section 3 above.